

Xv6-labs

```
khadejaq@DESKTOP-3E2P099: ~
make[1]: Leaving directory '/home/khadejaq/xv6-labs-2025'
== Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (3.5s)
== Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (0.8s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.2s)
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (1.0s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (1.3s)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: OK (1.1s)
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (0.7s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (1.0s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (0.8s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.0s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.4s)
== Test exec ==
$ make qemu-gdb
exec: OK (0.7s)
(Old xv6.out.find_exec failure log removed)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (1.0s)
```

```
khadejaq@DESKTOP-3E2P099: ~
$ make qemu-gdb
sleep, makes syscall: OK (1.2s)
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (1.0s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (1.6s)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: OK (1.7s)
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (1.1s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (1.2s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.1s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.0s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.2s)
== Test exec ==
$ make qemu-gdb
exec: OK (1.3s)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (1.5s)
== Test exec, recursive find ==
$ make qemu-gdb
exec, recursive find: OK (2.0s)
== Test time ==
time: FAIL
Cannot read time.txt
Score: 130/131
make: *** [Makefile:363: grade] Error 1
khadejaq@DESKTOP-3E2P099:~/xv6-labs-2025$
```

Time.c fails here but works when running separately and writes to txt file as well. Attached are screenshots

```
khadejaq@DESKTOP-3E2P096 x
+ v
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ time ls
.          1 1 1024
..         1 1 1024
README    2 2 2425
findtest.sh 2 3 91
sixfive.txt 2 4 19
cat        2 5 36080
echo       2 6 34952
forktest   2 7 16872
grep       2 8 39456
init       2 9 35432
kill       2 10 34896
ln         2 11 34720
ls         2 12 38040
mkdir      2 13 34976
rm         2 14 34952
sh         2 15 57952
stressfs   2 16 35824
usertests  2 17 184528
grind      2 18 50776
wc         2 19 37032
zombie     2 20 34304
logstress  2 21 36816
forphan    2 22 35712
dorphan    2 23 35152
sleep      2 24 34888
sixfive    2 25 36896
memdump    2 26 38360
find       2 27 40056
exec       2 28 40056
uptime     2 29 34608
time       2 30 36592
memdump    2 26 38360
console    3 32 0
3 ticks
$ |
```

```
khadejaq@DESKTOP-3E2P096 x
+ v
memdump    2 26 38360
console    3 32 0
3 ticks
$ cat time.txt
3
$ ls
.          1 1 1024
..         1 1 1024
README    2 2 2425
findtest.sh 2 3 91
sixfive.txt 2 4 19
cat        2 5 36080
echo       2 6 34952
forktest   2 7 16872
grep       2 8 39456
init       2 9 35432
kill       2 10 34896
ln         2 11 34720
ls         2 12 38040
mkdir      2 13 34976
rm         2 14 34952
sh         2 15 57952
stressfs   2 16 35824
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logstress  2 21 36816
forphan    2 22 35712
dorphan    2 23 35152
sleep      2 24 34888
sixfive    2 25 36896
memdump    2 26 38360
find       2 27 40056
exec       2 28 40056
uptime     2 29 34608
time       2 30 36592
memdump    2 26 38360
console    3 32 0
time.txt   2 33 2
$ |
```

Memdump, sixfive and sleep

```
khadejaq@DESKTOP-3E2P095 x + v
hart 1 starting
init: starting sh
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
850
1819438967
100
z
xyzzzy
Example 5:
hello
w
o
r
l
d
$ sixfive sixfive.txt
5
100
18
6
$ sleep 10
$ |
```

Find.c

```
khadejaq@DESKTOP-3E2P095 x + v
khadejaq@DESKTOP-3E2P095:~$ cd xv6-labs-2025
khadejaq@DESKTOP-3E2P095:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertest user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive us
er/_memdump user/_find user/_exec user/_uptime user/_time user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1214 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$
```

find.c with -exec

```
khadejaq@DESKTOP-3E2P096 x + v
zero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-bu
iltin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/find.o user/find.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_find user/find.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_find > user/find.asm
riscv64-linux-gnu-objdump -t user/_find | sed '1 /SYMBOL TABLE/d; s/ .* / /; /'$d' > user/find.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive us
er/_memdump user/_find user/_exec user/_uptime user/_time user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1217 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ find . README
./README
$ find . wc -exec echo hi
hi ./wc
$ sh < findtest.sh
$ $ $ $ $ hello
hello
hello
$ $ |
```